3FS

Deep Dive into DeepSeek — Chapter 30

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Foundation Books Project

The one rule of this chapter

A checkpoint write is not a checkpoint until a coherent generation can be identified, read, and restored.

- 3FS is a shared file interface plus stateless metadata services, built for SSD and RDMA paths, combining chunk placement with CRAQ-style replication.
- "The last byte landed" and "this generation is safe to restore" are different claims — the whole chapter is about the gap between them.
- The running example: a 10^{12}-parameter checkpoint, 1,024 writers, an hourly write interval, a ten-minute restore target.

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Goal: follow one checkpoint from namespace metadata through chunk placement, chain replication, client I/O, and target-local recovery to a restore-safe published generation.

Roadmap

Storage on the accelerator critical path

System boundary and namespace

Consistency and checkpoint publication

Target-local recovery and client I/O

Workloads, benchmarks, and operations

Storage on the accelerator critical path

3FS inside Fire-Flyer 2

Fire-Flyer 2 layer	Paper-reported role
Compute and training	About 1,250 nodes, 8 PCIe A100 GPUs each; HFReduce and HaiScale run collective communication and training.
Shared fabric	Two-zone InfiniBand fat-tree; service levels and static routing separate storage traffic from collective traffic.
3FS storage	180 dual-homed storage nodes, 16 NVMe SSDs and two 200-Gbps NICs each.

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Every 3FS read competes with HFReduce and NCCL on the same fabric — storage supply cannot be read off SSD bandwidth alone. [REPORTED: Fire-Flyer AI-HPC paper, arXiv:2408.14158]

Demand, supply, and the toy checkpoint

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$$\text{storage supply} = \min(\beta_{\text{ssd}}, \beta_{\text{net}}, \beta_{\text{service}}, \beta_{\text{metadata}}, \beta_{\text{client}})$$

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A storage claim needs the demand terms, the bottlenecked supply path, and the operating objective in the same record.

System boundary and namespace

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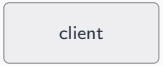

FUSE-facing namespace; I/O via calls or USRBIO

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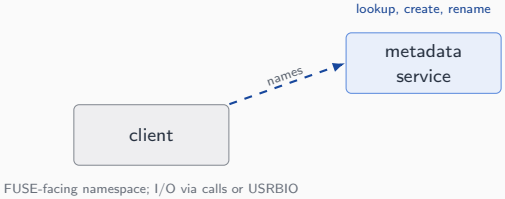

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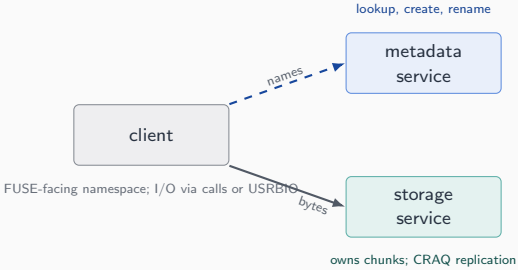

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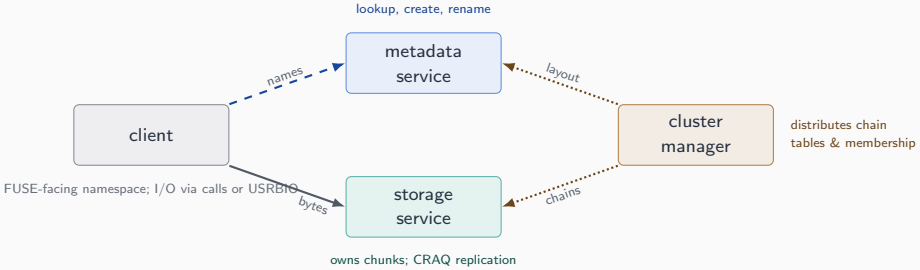

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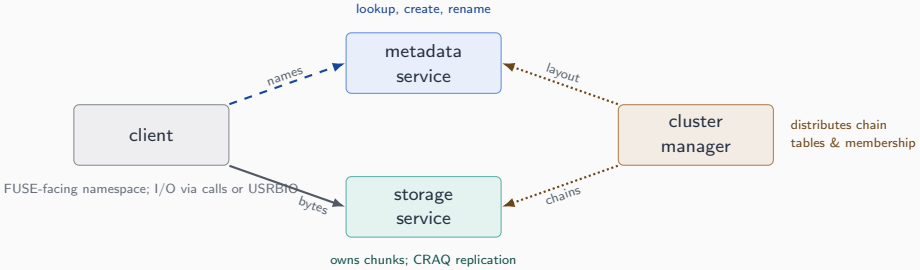

HFReduce, HaiScale, and the HAI Platform sit outside this boundary: they create traffic and decide completion, but 3FS owns the namespace, chunk state, and replica protocol.

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- A chunk is $\lceil B_{\text{worker}}/S_{\text{chunk}} \rceil$ pieces of one shard on an ordered chain; placement spreads a failed disk's redirected reads across many survivors, not just two.
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These three version counters need not advance together — a chunk-content retry cannot repair a stale table or a stale chain epoch.

Consistency and checkpoint publication

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"A replica has the new bytes" is weaker than "the normal write path completed."

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This state machine is a chapter-defined deployment rule built on 3FS semantics — not a literal 3FS API. The chunk transitions underneath it are source-backed.

Target-local recovery and client I/O

From concept to code: writing intent before commit

A storage target must not confuse newly written bytes with the metadata that makes them live.

from `3FS/src/storage/chunk_engine/src/types/chunk_meta.rs`

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`update_chunk` rejects a stale chain version, a committed/skipped update version, or a mismatched tag *before* publishing new metadata; only the RocksDB commit clears `uncommitted`. [INSPECTED: 3FS `src/storage/chunk_engine/src/core/engine.rs`]

FUSE and native USRBIO: one pathname, two data paths

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FUSE client	kernel FUSE → 3FS FUSE process	ordinary read/write calls

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Metadata setup is shared; payload submission and completion are not — so the two paths are not interchangeable in a benchmark report.

Workloads, benchmarks, and operations

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Fire-Flyer 2 reports a ∼5-minute checkpoint cadence and $> 10\,\mathrm{GiB/s}$ per node — source-reported context, not a reproduction of this toy ledger or a released-3FS-commit measurement. [REPORTED: Fire-Flyer AI-HPC paper, arXiv:2408.14158]

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This project has no 3FS deployment: peak-read, KV-cache, USRBIO, and checkpoint-restore reproduction are administratively closed, not measured zero or failed results. [OBSERVED: bounded repository-model and source-surface checks only]

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Bytes land on disk long before a generation is safe to restore.

Next: smallpond builds dependency-aware dataframe tasks on top of this same storage protocol (Chapter 31).